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TASK 2: Requirements Gathering

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1. STAKEHOLDER IDENTIFICATION

Project Scope

The primary objective of this project is to design and implement a mobile application that enables real-time collection and monitoring of Quality of Experience (QoE) data from mobile subscribers in Cameroon. This application is beneficial as it combines both the subjective feedback from users and the objective technical measurements from the mobile device, thereby bridging the gap between network performance and actual user experience.

Identity of Stakeholders

Stakeholders refer to anyone involved in, impacted by, or can directly influence the project. Below is a comprehensive list broken into categories of stakeholders.

1. Internal Stakeholders (Development Team)

- UI/UX designers: They are responsible for designing the mobile application's interface. Their role ensures that the mobile app is intuitive and provides a smooth user experience.
- Software developers: They are tasked with coding the mobile app (front-end) and server (back-end). They ensure that technical features like data collection, analytics, and security function properly.
- Project manager: Oversees the project timeline, resource allocation, and communication among the team.

2. External Stakeholders (End Users)

- Mobile network subscribers: These are individuals who use network services such as MTN and Orange in Cameroon. Their feedback is central to measuring Quality of Experience, making them a crucial part of the data collection process.

3. Key External Stakeholders (Mobile Network Providers)

- Orange Cameroon, MTN Cameroon, and Camtel: These mobile service providers will benefit from the QoE insights and are expected to collaborate by sharing necessary technical data and network information. Their cooperation is key to

validating performance metrics and integrating improvement strategies based on user experience.

4. **Tertiary Stakeholders (Indirect Influence):** These are the stakeholders that have a lower impact on the project but may still be affected by its outcomes.

- Regulatory authorities: They may be interested in the data outcomes for policy and regulatory purposes.
- Academic Researchers: They may make use of anonymized QoE data for studies on mobile network performance and user satisfaction in Cameroon.
- Investors or Sponsors: If applicable, they may provide funding and would require reports on project progress and impact.

Stakeholder Categorization and Prioritization

An engagement strategy refers to the methods and communication practices used to involve stakeholders in the project at the right level and frequency. The table below helps organize the different stakeholders by influence, priority, and engagement strategy.

Stakeholder	Category	Influence Level	Priority	Engagement Strategy
Software developers	Internal	High	High	Weekly meetings and clear task allocation
Project manager	Internal	High	High	Project planning and monitoring
UI/UX designers	Internal	High	High	Conduct user research to design user-friendly interfaces
End users	External	Medium	High	App onboarding and providing feedback reviews
Mobile network providers	Key External	High	High	Data-sharing agreements and providing the network service
Regulatory authorities	Tertiary	Low	Medium	Provide access to final reports and compliance updates
Academic researchers	Tertiary	Low	Low	Optional data-sharing agreements under license terms.

Table 1.1 : Stakeholder Analysis Table

2. REQUIREMENT GATHERING TECHNIQUES

Requirement gathering is a foundational phase in the software development lifecycle. It ensures that the final solution not only aligns with stakeholder expectations but also addresses the actual needs and challenges faced by users. For our **QoE Mobile App**, which aims to collect Quality of Experience (QoE) data from mobile network subscribers in Cameroon, we employed a range of requirement gathering techniques. These techniques helped us understand user behavior, technical expectations, and industry standards.

Below are the key techniques and how they will contribute to shaping our app:

Research Methods: Selection Process, Benefits & Activities

1. Document Analysis: We conducted a review of a research paper titled "**Modeling the Customer Experience Using Machine Learning for Optimizing the Performance of Telecommunications Networks**" by Binele Abana et al. (2024). This was done in order to have better insights into already established facts and insights.

Pros	Cons
Built on established scientific knowledge rather than assumptions	Some research might not apply to our specific context
Saved significant time in metric development	Time-intensive to thoroughly review literature

Table 2.1: Pros & Cons of Document Analysis

Activities Carried Out:

Key Insights Extracted:

1. The study emphasizes the gap between **Quality of Service (QoS)**—measurable metrics like latency, jitter, and throughput—and the **subjective Quality of Experience (QoE)** perceived by end-users.
2. It introduced the relevance of metrics such as **Mean Opinion Score (MOS)**, **signal strength**, and **call drop rates** as critical indicators of user satisfaction.

2. Stakeholder Interviews (Mobile Operators and Users)

We conducted direct interviews with both mobile operators like Orange and potential users of our app.

Pros	Cons
Built trust and relationships with key stakeholders	Time-intensive to conduct and analyze
Allowed us to explore unexpected topics as they emerged	Smaller sample size than surveys
Higher engagement compared to surveys	Potential interviewer bias affecting responses

Table 2.2: Pros & Cons of Stakeholder Interviews

Why this over alternatives: The personal nature of interviews helped us engage with stakeholders who might not respond to surveys. The conversations yielded rich insights and helped build relationships.

Activities Carried Out:

We engaged directly with industry stakeholders, especially representatives from major telecom providers like **Orange Cameroon**, to understand the operational side of user feedback collection.

Discussion Points:

1. Current metrics used to measure network performance and customer satisfaction.
2. Challenges in acquiring real-time user experience data.

3. Digital Surveys (Google Forms)

We used digital surveys as a complement to interviews. While users are often reluctant to fill out forms, this method allowed us to collect more responses quickly and from a broader audience.

Pros	Cons
Reached more people quicker	Lower response rates due to user reluctance
Provided quantifiable data we could analyze	Limited ability to probe deeper on interesting responses
Standardized format made comparing responses easier	Missed non-verbal cues and context
	Difficult to verify the quality of responses

Table 2.3 : Pros & Cons of Digital Surveys

Why this over alternatives: Google Forms gave us a quick way to gather quantifiable data at scale. We needed both the depth of interviews and the breadth of surveys to get a complete picture.

Activities Carried Out:

We designed and distributed structured questionnaires on Google Forms to mobile network subscribers

Content of Surveys:

1. Daily experiences with mobile services (e.g., network fluctuations, call clarity, internet reliability).
2. **Perception of app-based feedback** and concerns about privacy or complexity in using such tools.

4. Brainstorming Sessions

Our team needed to balance technical possibilities with user needs. Through brainstorming, we tackled tough questions about privacy, user motivation, and system integration.

Why this over alternatives: Collaborative brainstorming brought together diverse perspectives needed for this multifaceted project that spans technical, business, and user experience domains.

Pros	Cons
Combined insights from technical, design, and business perspectives	Could be dominated by louder voices or stronger personalities
Built team alignment and shared understanding	Required skilled facilitation to stay productive
Identified potential issues before they became problems	Time-consuming to organize with the whole team
	Harder to document than more structured methods

Table 2.4: Pros & Cons of Brainstorming Sessions

Activities Carried Out:

We held multiple brainstorming sessions with our team. These sessions were essential in generating innovative ideas and structuring our approach.

Key Activities:

1. Ideation around questionnaire formats.
2. Identifying relevant stakeholders
3. Establishing data cleaning procedure

3. DATA GATHERING

Introduction

The primary focus of the data gathering process was on collecting feedback about Quality of Experience (QoE) from users of the three major mobile network providers: MTN, Orange, and Camtel. The goal was to assess user satisfaction levels, identify common frustrations, and gauge interest in a potential mobile application for ongoing QoS feedback.

Data Collection Methodology

Collection Approach

A direct interview approach with potential users was employed, using a structured questionnaire. This allowed us to gather both quantitative metrics (ratings, frequencies) and qualitative insights (opinions, frustrations, behaviors). The interviews were conducted in person, and responses were recorded on paper and then transferred into an Excel spreadsheet and saved as CSVs.

Sample Characteristics

The study involved 60 respondents, primarily through convenience sampling - approaching people in various locations. The demographic breakdown shows:

- Age distribution: Predominantly young adults (18-25 years), with a smaller representation of users in the 26-35 age bracket
- Network providers: Users of MTN, Orange, and Camtel, with many respondents using multiple providers
- Geographic focus: The location from which data was collected was the township of Buea.

Data Types Collected

The study gathered a balanced mix of:

Quantitative data:

- Demographic information (age groups)

- Usage frequency metrics (internet, voice calling, SMS usage)
- Satisfaction ratings (on a scale of 1-5)
- Problem frequency measurements (how often users experience poor network)

Qualitative data:

- User behaviors when experiencing network issues
- Main frustrations with current service providers
- Reasons behind satisfaction/dissatisfaction ratings
- Factors that might discourage app usage
- Comfort levels with background data collection

Questionnaire Design

The interview questionnaire was designed to capture a comprehensive picture of the mobile user experience. Key question areas included:

1. **Usage patterns:** "How frequently do you use internet/voice calling/SMS services?"
2. **Problem assessment:** "How often do you experience poor network connectivity?"
3. **Coping mechanisms:** "What do you typically do when you experience network problems?"
4. **Satisfaction measurement:** "On a scale of 1-5, how satisfied are you with your current network provider(s)?"
5. **Provider history:** "Have you switched network providers in the past?"
6. **Key pain points:** "What is your main frustration with your network provider?"
7. **App reception:** "Would you be willing to rate your network experience using a mobile app?"
8. **App concerns:** "What would make you hesitate to use such an app?"
9. **Incentive assessment:** "Would you be more motivated if the app offered small rewards for giving feedback?"
10. **Privacy comfort:** "How comfortable are you with an app collecting network information in the background?"

Service Provider Input

To complement user perspectives, we also gathered input from mobile service providers about their current QoE data collection methods:

- **Current collection channels:** Customer service calls, WhatsApp communications, word of mouth feedback, and walk-in complaints
- **Existing tools:** Internal apps used within organizations for data collection, but no dedicated user-facing applications
- **Challenges identified:** Concerns about data validity (possibly false information) and timeliness of user feedback
- **Key metrics tracked:** Voice call clarity, internet speed, and interaction timestamps

Key Findings from Data Collection

User Satisfaction and Experiences:

1. **Overall satisfaction is low:** The majority of respondents rated their satisfaction at 1 or 2 on a 4-point scale, indicating widespread dissatisfaction with current service quality.
2. **Common frustrations:** Users frequently cited these key issues:
 - Unreliable or unstable connections
 - Slow internet speeds
 - Expensive services
 - Rapid data consumption
3. **Frequent connectivity issues:** Most respondents reported experiencing poor network connectivity either "Often" or "Daily," suggesting persistent service problems.
4. **Coping behaviors:** When facing connectivity issues, users typically:
 - Toggle airplane mode on and off
 - Switch between different network providers
 - Wait for service to resume
 - Change physical location

Data Quality Assessment

Strengths of the Dataset

- Rich mix of quantitative metrics and qualitative insights
- Good coverage across the three major network providers
- Captured both user frustrations and practical coping behaviors
- Included provider perspectives to complement user feedback

Limitations

- The age demographic skewed heavily toward younger users (18-25)
- Some missing data points in the CSV file (blank fields)
- Limited geographic information to analyze regional variations
- Convenience sampling may not be fully representative of all user groups

4. DATA CLEANING

Background and Context

The effectiveness of any data-driven application lies in the quality of its data. As part of the development of our network-feedback app, our team initiated a data-cleaning process to ensure the reliability and usability of the feedback collected from diverse stakeholders.

Initial Data Review

The initial review revealed several common issues inherent in community-sourced data, namely:

- Incomplete Entries
- Irrelevant or Off-topic Responses
- Duplicate records
- Blank Submissions
- Outliers (e.g., Network score of 100 on a scale of 1-5)
- Unstructured Text (For Free text inputs)
- Mixed Languages and Slangs
- Spelling and Typographical Errors

Data Entry

To prepare the collected feedback for analysis, all responses were systematically entered into a structured dataset. Given the diverse methods of data collection, different entry techniques were employed to ensure uniformity across sources:

- **Manual Entry:** Feedback gathered through **in-person interviews** with community members (especially elders and youths) was manually keyed into an Excel spreadsheet by team members following a predefined format.
- **Digital Export:** Responses obtained via online forms, such as **Google Forms** distributed to ISP administrators and tech-savvy youths, were exported directly into CSV format for

ease of integration.

- **Data Consolidation:** All datasets, whether manually input or digitally sourced, were **reviewed and merged** into a single working file to prepare for the cleaning process.

Data Cleaning Process

After consolidating the collected data, a structured cleaning process was undertaken to enhance accuracy, consistency, and readiness for analysis. The raw dataset, as expected, exhibited several issues, such as missing values, inconsistent formatting, typographical errors, and duplicates.

Tools and Libraries Used:

- **Python** (mainly for automated and scalable cleaning)
 - a. **Pandas** – for handling missing data, filtering, and transforming the dataset.
 - b. **NumPy** – for handling numerical inconsistencies and missing values.
- **Microsoft Excel** – used for manual inspection, quick visual corrections, and verifying changes.

Cleaning Steps Undertaken:

- **Handling Missing Values**
 - a. Used `df.isnull().sum()` to identify missing fields.
 - b. Used `isinstance(x, str) else False).sum()` to check for missing strings.
 - c. Filled non-critical fields like frequency_columns' text entries with "Unspecified" using `fillna()`

```

#checking for missing values
print(f"\nMissing Values: {df.isnull().sum()}")

#checking for empty strings
print("\nEmpty strings in set:")
empty_strings = df.map(lambda x: x== '' if isinstance(x, str) else False).sum()
display(empty_strings[empty_strings > 0])

#Filling nan values with 'Unspecified'
df_clean['Network_Provider'].fillna('Unspecified', inplace =True)
print(df_clean['Network_Provider'].head())

#Now to clean the frequency columns by filling in their missing values
frequency_columns = ['Internet_Usage_Frequency', 'Voice_Calling_Frequency', 'SMS_Usage_Frequency']
for col in frequency_columns:
    if df_clean[col].isnull().sum() > 0:
        #fill with 'unspecified'
        df_clean[col] = df[col].fillna('Unspecified')

```

Missing Values: Respondent_ID 6
Age_Group 6
Network_Provider 30
Internet_Usage_Frequency 23
Voice_Calling_Frequency 44
SMS_Usage_Frequency 44
Poor_Network_Frequency 6
Response_To_Network_Problems 6
Provider_Switch_History 6
Main_Frustration 6
Satisfaction_Rating 6
Satisfaction_Reason 6
App_Affinity 6
Hesitation_Factors 8
Reward_Motivation 6
Comfort_With_Background_Info_Collection 6
dtype: int64

Empty strings in set:
Series([], dtype: int64)

df_clean['Network_Provider'].head()

Image 4.1: Python Code to automate checking and management of missing fields

- **Standardizing Text Entries**

- Converted Satisfaction Ratings to Numeric Input using
`pd.to_numeric(df_clean['Satisfaction_Rating'])`

```

#converting staisfaction rating to numeric if it's not
if df_clean['Satisfaction_Rating'].dtype == 'object':
    df_clean['Satisfaction_Rating'] = pd.to_numeric(df_clean['Satisfaction_Rating'])

#Now standardizing text columns to ensure consistency
boolean_columns = ['App_Affinity', 'Hesitation_Factors', 'Reward_Motivation', 'Comfort_With_Background_Info_Collection']
for col in boolean_columns:
    df_clean[col] = df_clean[col].str.lower if df_clean[col].dtype == 'object' else df_clean

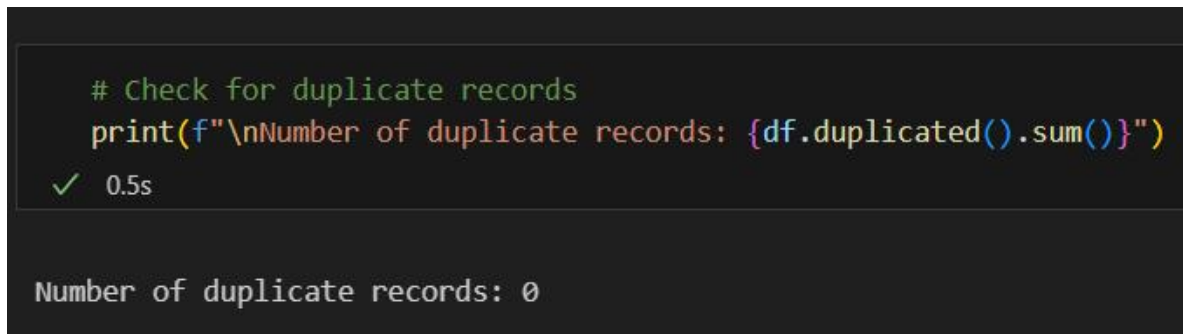
```

Image 4.2: Python code to standardize dataset

- Standardized text columns to ensure consistency for text inputs

- **Removing Duplicates**

- a. Used `df.duplicated().sum()` to identify duplicate records.



```
# Check for duplicate records
print(f"\nNumber of duplicate records: {df.duplicated().sum()}")
```

✓ 0.5s

Number of duplicate records: 0

Image 4.3: Python code to check duplicate records

This multi-tool approach allowed the team to clean the dataset efficiently while ensuring flexibility in reviewing both automated and manually corrected entries. The cleaned dataset provided a strong foundation for downstream analysis and insights.

Results of Cleaning:

The initial Excel dataset contained entries, many of which included missing values, inconsistent location names, and duplicate feedback. After applying cleaning procedures using Python and Excel, the dataset was refined to standardized records, as shown in the images below.

This cleaned version now ensures accurate analysis and reliable insights for the network-feedback app.

Report ID	Age_Group	Network_Provider	Internet_Usage_Frequency	Voice_Calling_Frequency	SMS_Messaging_Frequency	Poor_Network_Frequency	Response_Time_Problem_Solving	Main_Issue	Frustration_Level	Satisfaction_Rating	Avg_Air_Usage_Hourly	Health_Factors	Reward_Motivations
1-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Often	Wait	No	Unreliable	3	Location	Bound	"bound bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
2-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Daily	Wait	Yes	Unreliable	1	Bound	Bound	"bound bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
3-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Daily	Wait	Yes	Unreliable	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
4-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Often	Switch	Yes	Unreliable	3	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
5-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Often	Wait	Yes	Unreliable	3	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
6-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Often	Bound	Airplane mode	Expensive	1	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
7-10-25	Unspecified	Daily	Unspecified	Unspecified	Unspecified	Daily	Switch	Yes	Unreliable	3	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
8-10-25	Orange	Daily	Unspecified	Unspecified	Unspecified	Often	Wait	Yes	Expensive	4	Feister	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
9-10-25	Orange	Daily	Unspecified	Unspecified	Unspecified	Yes	Mechanical	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
10-10-25	MTN	Daily	Unspecified	Unspecified	Unspecified	Often	Airplane mode	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
11-10-25	MTN	Daily	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	Yes	Slow	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
12-10-25	MTN	Daily	Unspecified	Unspecified	Unspecified	Daily	Switch	Yes	Slow	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
13-10-25	MTN	Unspecified	Unspecified	Unspecified	Unspecified	No	Switch	Yes	Inexplicable	3	Acceptable	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
14-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	No	Switch	Yes	Slow	3	Null	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
15-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Switch	Yes	Expensive	1	Null	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
16-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	No	Expensive	1	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
17-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Switch	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
18-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	Yes	Expensive	3	Feister	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
19-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Switch	Yes	Slow	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
20-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Wait	Yes	Slow	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
21-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Often	Switch	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
22-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Often	Airplane mode	Yes	Unreliable	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
23-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Often	Airplane mode	No	Slow	1	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
24-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Often	Airplane mode	Yes	Slow	1	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
25-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
26-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
27-10-25	Unspecified	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"
28-10-25	MTN, Orange	Unspecified	Unspecified	Unspecified	Unspecified	Daily	Airplane mode	Yes	Expensive	2	Poor network	Bound	"bound bound m-bound m-bound method StringMethods.lower of <android.content.Context>StringMethods object at 0x0000000BFE6F6580"</

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5. USER RELUCTANCE ASSESSMENT

This section examines the factors that contribute to user reluctance in adopting and engaging with a mobile application designed to collect network quality feedback from subscribers. The findings presented herein are derived from survey and interview data collected from mobile network subscribers.

Definitions and Key Terms

User Reluctance: The hesitation, unwillingness, or resistance exhibited by potential users toward adopting a new technology solution, in this case, a network feedback application.

Privacy Concerns: Apprehensions related to the collection, storage, usage, and potential sharing of personal data gathered through the application.

Technical Barriers: Obstacles stemming from the application's design, functionality, or technical requirements that impede user adoption or engagement.

Trust Deficit: The lack of confidence in service providers' intentions, reliability, or responsiveness to user feedback.

Resource Consumption: The utilization of device resources such as data, battery life, or storage space by the application.

User Adoption: The process by which potential users become actual users of the application, moving from awareness to installation and regular engagement.

User Engagement: The level and quality of user interaction with the application after initial adoption.

Incentivization: The implementation of rewards or benefits designed to motivate user participation and continued engagement such as free airtime or data

Categories of User Reluctance

Privacy Concerns

Survey respondents expressed significant reservations regarding data collection practices, particularly concerning location tracking and background monitoring. These concerns stem from:

- Uncertainty about the extent and purpose of data collection
- Concerns about potential misuse of personal information
- Lack of clarity regarding data storage and security measures

Representative survey responses include:

- "Not too comfortable" with background data collection
- "I'm comfortable if the extent of the info collected is just signal strength but not location."
- "Not comfortable at all" with background monitoring

Technical and Design Barriers

Potential users demonstrated aversion to applications that require significant effort to use or that create friction through complex interfaces. Main concerns include:

- Complex or lengthy forms requiring extensive user input
- Multiple-step processes that consume time and attention
- Excessive notifications that disrupt normal device usage
- Complicated user interfaces requiring technical knowledge

Representative survey responses include:

- "Lengthy forms" (mentioned by multiple respondents)
- "Too many requirements and steps"
- "Too many notifications"
- "Too much work"

Trust Deficit with Network Operators

Survey data reveals a substantial lack of confidence in mobile network operators' intentions and responsiveness. This trust deficit manifests as:

- Skepticism about operators' willingness to act on feedback
- Doubt regarding the genuine interest of operators in user experience
- Negative prior experiences create resistance to further engagement
- Perception that feedback collection is merely performative

Representative survey responses include:

- 52% responded "No" when asked if operators care about users
- 38% responded "Maybe" to the same question
- Comments such as "My God will judge you!" and "Man... your network sucks"
- Average satisfaction rating of 2.1 on a 5-point scale

Do you feel like mobile operators care about your experience as a customer?
21 responses

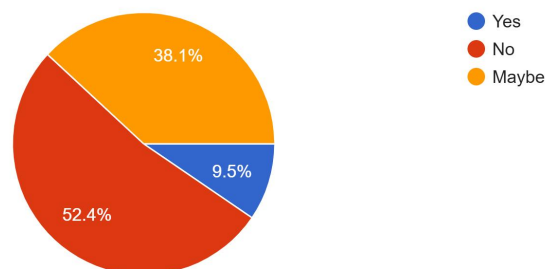


Image 5.1: User reviews on perceived operator care

Resource Consumption Concerns

Given the reported network reliability issues, concerns about the application's impact on limited device resources were prominent:

- Data usage for uploading feedback and metrics
- Battery drain from background monitoring processes
- Large application size consuming limited device storage

Representative survey responses include:

- Multiple mentions of "data usage concerns"
- "You'll be using my battery."
- Concerns about application "weight"

Mitigation Strategies for Reducing User Reluctance

Addressing Privacy Concerns

Strategy	Implementation Approach	Expected Impact
Transparent data policies	Implement clear, non-technical explanations of all data collection	Reduces uncertainty and builds trust
Granular permissions	Allow selective enabling/disabling of specific monitoring types	Increases user control and comfort
Anonymous reporting	Separate identity from network performance data	Reduces concern about personal targeting

Table 5.1: Possible strategies to mitigate privacy concerns

Overcoming Technical Barriers

Strategy	Implementation Approach	Expected Impact
Minimalist interface	Limiting UI to essential functions with minimal interaction steps	Reduces cognitive load and user effort

Table 5.2: Possible strategies to mitigate technical concerns

Building Trust and Credibility

Strategy	Implementation Approach	Expected Impact
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Public results	Publish anonymized, aggregated feedback visible to all users	Creates transparency and accountability
Community partnerships	Collaborate with trusted local organizations	Transfers established trust to the application
Operator commitments	Feature public pledges from operators to address identified issues	Sets expectations for response

Table 5.3: Possible strategies to mitigate credibility concerns

Addressing Resource Consumption

Strategy	Implementation Approach	Expected Impact
Data-saver mode	Collect and upload data only when connected to WiFi	Eliminates data usage concerns
Resource monitoring	Display clear metrics on the application's resource usage	Creates transparency about impact
Offline functionality	Enable feedback collection without an active connection	Accommodates intermittent connectivity
Lightweight design	Keep application under 10MB with optimized resource usage	Addresses storage concerns

Table 5.4: Possible strategies to mitigate resource consumption concerns

Incentivization Approaches

Survey data indicates that 76% of respondents would be more motivated if incentives were offered. Recommended approaches include:

Strategy	Implementation Approach	Expected Impact
Micro-rewards	Provide small airtime credits for specific feedback actions	Creates immediate tangible value

Recognition system	Implement community acknowledgment for contributors	Appeals to social motivations
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Table 5.5: Possible incentivisation approaches

Would you be more motivated if the app offered small rewards (like airtime or data bonuses) for giving feedback?
 21 responses

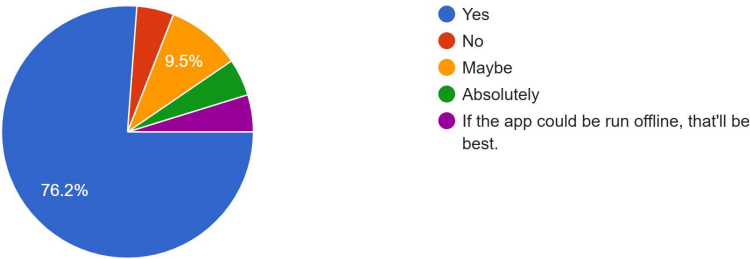


Image 5.2: User feedback on incentivisation

The strategies outlined provide a framework for reducing reluctance through thoughtful application design, transparent communication, trust-building measures, and appropriate incentivisation. By implementing these approaches, internal stakeholders can create a feedback ecosystem that overcomes initial resistance and establishes sustainable user engagement.

Conclusion

The data-gathering process successfully captured valuable insights into mobile network user experiences in Cameroon. The approach balanced quantitative measurements with qualitative feedback, creating a comprehensive picture of user satisfaction, frustrations, behaviors, and receptiveness to a potential QoE feedback application.

The findings reveal significant dissatisfaction with current service quality, suggesting a clear need for improvements.

The contrasting perspectives from users and service providers highlight an information gap that could be bridged through more systematic feedback collection. While providers currently rely on passive and ad-hoc methods, users are experiencing frequent problems that may not be fully captured through these channels.

These insights provide a solid foundation for the design and development of a mobile application focused on QoE feedback collection, addressing the needs of both users and service providers in the Cameroonian mobile telecommunications landscape.

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